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16 JULY 1962

SUPERSEDING
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MILITARY SPECIFICATION

MOUNT, GUN; MOUNT, HOWITZER

1. SCOPE

1.1 **Scope.** This specification covers two types of gun mounts, Hydrospring and Hydro-pneumatic to be mounted in combat vehicles.

1.2 **Classification.** Mounts shall be furnished in the following types and classes (see 6.2) .

Type I—Hydrospring, this type of system(s) is usually limited to light artillery and short recoil distances (see 6.4).

Type II—Hydro-pneumatic, this type of system(s) can be adapted to either light or heavy artillery (see 6.4).

Class I-Concentric recoil mechanism, (concentric spring type), this class has one spring, concentric with the gun tube.

Class 2-Concentric recoil mechanism, (multiple cylinder type),this class has a multicylinder recoil mechanism. The cylinders are usually placed about the exterior of the cradle. The number of cylinders are usually one or more, depending upon the function each type is required to perform.

Class 3—Concentric recoil mechanism with separate counter-recoil assembly, this class has the counter-recoil mechanism independent of the recoil brake.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

BB-N-411 —Nitrogen
U-P-271 —Paper, Wrapping. waterproofed Kraft
UU-T-81 —Tags, Shipping and Stock.
PPP-T-60 —Tape; Pressure-sensitive-Adhesive Waterproof for Packaging and Sealing.

MILITARY

JAN-P-100 —Packaging and Packing for Overseas Shipment, General Specification.
MIL-C-104 Crates,Wood; Lumber and Plywood Sheathed Nailed and Bolted.
MIL-P-116 -Preservation, Methods of.
MIL-B-121 -Barrier Material, Greaseproofed.
MIL-H-6083 -Hydraulic Fluid, Petroleum Base, Preservative.
MIL-G-10924 -Grease, Automotive and Artillery.

FSC-1010, 1015, 1025

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MIL-P-11443 —Packaging and Packing for Shipment and Storage of Spare Parts for Military Vehicles.
MIL-V-13811 —Varnish, Waterproofing, Electrical, Ignition.

STANDARDS

MILITARY

MIL-STD-105 -Sampling Procedures and Tables for inspection by Attributes.
MIL-STD-129 —Marking for Shipment and Storage.
MIL-STD-180 —Identification Marking of U.S. Military Property.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 *Preproduction.* When specified (see 6.2), a preproduction model (s) shall be produced prior to the manufacture or fabrication of gun mounts in quantity. The complete gun mounts shall be submitted for inspection and tests to determine whether it has been constructed in accordance with 3.4, and whether it meets the performance requirements specified herein. Gun mount submitted shall be fully representative of gun mount proposed to be supplied by the manufacturer from production facilities and tooling as contemplated under the supply contract.

3.2 **First production gun mount and recoil mechanisms assembly.** Three or more of the first production gun mount and recoil mechnism assemblies produced from production tooling shall be furnished for all inspection and tests specified herein.

3.3 **Materials**

3.3.1 *Qualified products.* Gun mount manufacturers shall be responsible for ascertaining that all assemblies and component parts entering into the construction of each gun mount, which require product qualification in accordance with a specification, are listed on respective qualified products lists (QPL) or have been approved for inclusion on such list. Manufacturer's inspection record shall specifically list such components, name of supplier (s), and number and date of applicable QPL.

3.4 **Construction.** The gun mount, components; assemblies; and subassemblies shall be constructed and assembled into a complete assembly in accordance with drawings specifications, and standards listed or referred to in the EPL applicable to type and class specified in 1.2, or as applicable to newly developed classes. All parts shall be identified in accordance with Standard MIL-STD-130.

3.4.1 *Hydraulic system.* The initial and subsequent filling of the hydraulic system shall be with hydraulic oil, filtered through a 5-micron filter, conforming to Specification MIL-H-6083. Prior to and after gymnastication at contractor's plant the hydraulic system shall be free of all metallic or abrasive particles that may cause damage to the inner surfaces of the operating mechanisms, or malfunction of the system. These particles shall not be larger than that specified on the applicable drawing.

3.4.2 *Polished surfaces cleanliness.* All finish machined parts shall be free from perspiration residues, from final machining, until final assembly. Additionally all parts and components entering into the assembled mount shall be free of all other foreign matter after assembly.

3.4.2.1 *Parts and components modification.* No part or component shall be altered or hand fitted during assembly, unless the modification is or has been authorized by an approved drawing change.

3.4.3 *Air, gas, and accumulator system.* Air, gas, and accumulator system including lines, couplings, valves, pumps, reservoirs, chambers, filters and other parts shall be clean and free of all foreign matter. The hydroscopic content of the air used to blow lines shall not exceed 1 percent of the total volume of air used. Gas systems shall be

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filled, bled and refilled until all air and moisture is eliminated.

3.4.4 *Cylinders.* The finished cylinders of the recoil mechanism shall withstand pressures specified on applicable drawings without damage and permanent deformation.

3.4.5 *Piston, piston rods, and nuts or collars.* The finished piston, piston rods, and nuts or collars shall withstand the pull stress specified on applicable drawings, without permanent set or failure of the screw threads.

3.4.6 *Lubrication.* Lubricant conforming to Specification MIL-G-I0924, shall be used in lubrication of the assembled mechanism.

3.5 **Performance.** The cannon, gun mount and recoil mechanism, completel y assembled and lubricated shall operate without chatter or binding in all ranges of elevation and depression in ambient air temperatures of minus 25° to plus 115° F., without special aids under all operating conditions and as specified herein. When specified (see 6.2) special aids shall be provided for temperatures of minus 65° and plus 125° F.

3.5.1 *Counter-recoil velocity.* Recoil cylinder oil pressure versus time, recoil travel and velocity versus time (counter recoil), shall be within limits specified on applicable drawings. Counter-recoil velocity at 1 inch from the complete in-battery position, shall not exceed that specified on the applicable drawing.

3.5.2 *Gymnastication and static recoil cylinder pressurization.*

3.5.2.1 *Gymnastication.* The assembly consisting of the cannon or simulated equal, gun mount, recoil mechanism, replenisher or reservoir attached, torque bracket (s) assembly attached, and breech in operating cam assembly complete, shall be gynmasticated at the contractor's plant at elevations specified in the appropriate column of table I.

3.5.2.1.1 *Return cycle time.* During each gymnastication the return cycle (counter recoil) time shall be as specified on applicable drawings.

3.5.2.2 *Static pressure application hydrospring type.* The recoil mechanism assembly with cannon in place shall withstand hydraulic pressure as specified on applicable drawings for not less than 5 minutes and as applicable to specific gun mount and recoil mechanism assembly.

Caution: Extreme caution must be exercised during pressurization of recoil mechanism.

3.5.2.3 *Charging and filling hydro-pneumatic type.* Charging and filling the nitrogen chamber with gas and oil shall be in accordance with applicable drawings for a specific assembly, or as authorised by the procuring activity. Nitrogen and recoil shall conform to Specification BB-N-411, and MIL-H-6083 respectively, or as specified by the proccuring activity. Prior to charging the nitrogen chamber to full capacity the chamber shall be pressurized with nitrogen to a pressure of 50 pounds per square inch (psi) and released until free of trapped air and moisture.

3.5.3 *Provisions for proof-firing.* The complete gun mount and recoil mechanism assembly including all firing components shall withstand proof-firing as specified herein without damage, when mounted and proof-fired from a vehicle or simulated vehicular application (see 6.6).

3.5.3.1 *First production gun mount and recoil mechanisms assembly.* First production gun mount and recoil mechanism assembly shall be subjected to proof-firing from a vehicular application, or as specified (see 6.6). The full automatic assembly shall be subjected to firing bursts of three rounds or more as may be needed to fully prove-out assembly requirements. All proof-firing shall be as specified in table I.

3.5.3.1.1 *Climatic conditions.* In addition to proof-firing specified in 3.5.8.1 and table I, rounds 1 through 15 shall be repeated in each of the following climatic phases:

(a) Phase I at plus 70° F.
(b) Phase II at plus 125° F.
(c) Phase III at minus 65° F.

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TABLE I. *First production gun mount and recoil mechanism assembly proof-firing.*

Round No.	Rounds to be fired	Bursts for FA	Percent of maximum rated powder pressure				Firing elevation degrees			
			Tank	SPW	SS	FA	Tank	SPW	Single shot	Full automatic
1	1	.....	50	50	50	100	0	0	0	0
2	1	.....	75	75	75	100	0	0	0	0
3	1	.....	100	100	100	100	0	0	0	0
4	1	.....	115	115	100	100	0	0	45	20
5	1	.....	115	115	115	100	10	0	45	45
6	1	.....	100	100	115	100	10	45**	45	45
7	1	.....	115	115	100	100	20	45**	Max	45
8	1	B*	100	100	115	100	35	Max	Max	45
9	1	B*	115	115	100	100	35	Max	Max	45
10	1	.....	115	115	.....	100	35	Max	.....	45
11	1	.....	100	100	.....	115	0	0	.....	Max
12	1	.....	.....	.....	.....	115	.....	.....	.....	Max
13	1	.....	.....	.....	.....	100	.....	.....	.....	Max
14	.....	B*	.....	.....	.....	100	.....	.....	.....	Max
15	1	.....	.....	.....	.....	100	.....	.....	.....	Max

Notes. 1. Full automatic. (FA)
2. *Bursts for the full automatic only (B)
3. Self-propelled weapons. (SPW)
4. Single shot. (SS)
5. **For variable recoil mechanisms these elevations will be as determined by the cross over between long and short recoil. These angles will be as specified on applicable drawings. The intent is that the upper elevation limit for minimum recoil and the lower limit for maximum recoil be checked.

3.5.3.1.2 *First production gun mount and recoil mechanism assembly instrumentation.* All first production gun mount and recoil mechanism assembly shall be instrumented to determine recoil cylinder oil pressure versus tiem, recoil travel and velocity versus time. The limits in performance shall be the same as specified in 3.5.4.

3.5.4 *Production gun mount assemblies.* Each production gun mount submitted for acceptance shall be proof-fired as specified in Table II in prevailing ambient air temperature.

TABLE II. *Production proof-firing.*

Round No.	Rounds to be fired	Percent of maximum rated powder pressure				Firing elevation degrees			
		Tank	SPW	SS	FA	Tank	SPW	Single shot	Full automatic
1	1	50	50	50	100	0	0	0	0
2	1	75	75	75	100	0	0	0	0
3	1	100	100	100	100	0	0	0	0
4	1	115	115	100	100	0	0	10	10
5	1	115	115	115	100	20	*0	20	20
6	1	115	115	115	115	35	*45	45	45
7	1	110	100	115	115	.....	Max	Max	Max
8	.....	.....	.....	100	100	0	0	Max	Max
9	.....	.....	.....	100	100	.....	.....	Max	Max
10	.....	.....	.....	100	100	.....	.....	10	10

Notes. 1. Full Automatic. (FA)
2. Self-Propelled Weapons. (SPW)
3. Single Shot. (SS)
4. Single Shot. (SS)
5. **For variable recoil mechanisms these elevations will be as determined by the cross over between long and short recoil. These angles will be as specified on applicable drawings. The intent is that the upper elevation limit for minimum recoil and the lower limit for maximum recoil be checked.

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3.5.4.1 *Breech operation (sliding* breech *type).* During counter-recoil action, the breech operating cam shall operate in unison with breech mechanism to produce ejection of empty cases clear of the cannon and latch breechblock in open position. The velocity of ejected cases shall be specified on applicable drawings.

3.5.4.2 *Automatic breech mechanism operation.* Automatic breech mechanism shall operate in such a manner so as to provide complete access for loading of the next round of ammunition. The automatic breech closing mechanism shall completely close and lock the breech to provide complete obturation about the breech area.

3.5.4.3 *Adapters and collars.* All adapters, collars, and fastening devices (used to fasten recoil mechanisms, pistons, rods, etc. to the cannon) shall be tightened to torque specified in applicable drawings.

3.5.5 *Leakage.* During and immediately after the completion of proof-firing the first of contract and production gun mounts, the oil leakage from recoil mechanism shall not be more than 15 drops in a 3-minute period.

3.5.6 *Chamber pressure.* The chamber pressure and muzzle velocity shall be in accordance with applicable cannon ballistic drawing.

3.5.7 *Recoil travel.* When proof-firing the gun mount with all 100 and 115 percent charges, the length of cannon recoil travel shall be in accordance with applicable drawings.

3.5.7.1 *Recoil cycle time.* When proof-firing the gun mount at specified degrees of elevation or depression as required, the recoil cycle time shall be as specified on applicable drawings or by the procuring activity.

3.5.8 *Operation, complete recoil cycle. Operation* during the complete recoil cycle shall be smooth and continuous within the limits established herein. Conditions such as erratic motion surge, hang-up during recoil action rebound on return to battery (counter-recoil) shall not be permitted.

3.5.9 *Safety devices.* The safety device provialed shall securely lock the firing mechanisms in “safe” position.

3.5.10 *Electrical firing system.* The complete gun mount electrical circuit resistance shall not be greater than that specified on the applicable drawing, nor shall it show evidence of short circuiting.

3.5.10.1 *Electrical component.* All electrical components shall be capable of firing the ammunition in the cannon. The ammunition primer, cannon and cannon electrical system integrated with the gun mount electrical system, and external power source shall be considered as a complete gun firing system.

3.5.11 *Linkage assembly for percussion firing.* Linkage assemblies for cannon percussion firing actuators shall operate smoothly and efficiently. Setting of linkage headspace shall be in accordance with applicable drawings.

3.6 *Workmanship.* Workmanship shall be of a quality to assure gun mount and components free of defects (chip, burs, scratches and stains on finished surfaces, improper machining, and out of tolerance), due to improper fabrication, machining and handling methods.

4. QUALITY ASSURANCE PROVISIONS

4.1 *Contractor inspection.* The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *Contractor's quality control system.* The contractor shall provide and maintain

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an effective inspection and Quality control system acceptable to the Government covering the supplies under the contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The contractor will not be restricted to the inspection station or to the method of inspection listed provided that an equivalent control is included in the approved quality control procedure. The contractor shall notify the Government of and obtain approval for any change to the written procedure that might affect the degree of control required by this specification or other applicable documents referenced therein.

4.2 **Government verification.** All quality control operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written inspection plan are being properly applied, and (b) Government product inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have an effect upon the quality of the product, will be immediately called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.3 **Preproduction.**

4.3.1 **Preproduction inspection.** A complete preproduction gun mount assembly when furnished (see 3.1) shall be inspected to ascertain conformance with requirements of this specification. Inspection shall be performed at a location approved by the Government. The mount selected shall be completely dismantled and inspected 100 percent for conformance to all drawing requirements.

4.4 **Qualified products.** The contractor inspection records shall be checked to determine conformance to 3.3.1.

4.5 **Hydraulic system.** To determine conformance to 3.4.1, after gymnastication at the contractor's plant the gun mount hydraulic oil shall be drained and filtered through appropriate filter. Evidence of contamination of the hydraulic oil by this filtering action shall be cause for rejection of the recoil mechanism.

4.6 **Materiel assembly check.** To determine conformance to 3.4.2, the assembled materiel shall be examined for finished surface defects, cleanliness of assembled materiel, and for security of fastening devices.

4.7 **Air, gas, and accumulator system check.** To determine conformance to 3.4.3, the systems shall be examined for cleanliness, air, and moisture.

4.7.1 *Cylinder check.* To determine conformance to 3.4.4, cylinders shall be checked for damage and permanent set. Pressures shall be checked with adequate gages, having been periodically calibrated for accuracy.

4.7.2 *Piston, piston rods, and nuts or collar check.* To determine conformance to *3.4.5,* piston, piston rods, and nuts or collars shall be examined for permanent set or damaged threads.

4.7.3 *Lubrication* check. To determine conformance to 3.4.6, assembly shall be checked for proper lubrication.

4.8 **Performance tests.** To determine conformance to 3.5, the complete assembly shall be subjected to tests specified herein.

4.8.1 *First production gun mount and recoil mechanism assembly proof-firing* test. To determine conformance to 3.2, 3.5.1 and 3.5.3 the three first production gun mount and recoil mechanism assemblies shall be submitted to a Government proving ground for proof-firing testing. Failure of any of the three gun mounts to conform with requirements specified herein shall be cause for refusal by the Government to conduct more test or retests until corrective measures satisfactory to the Government have

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been taken. After corrections have been made by the contractor, another lot of three gun mounts shall be submitted for test. If failure occurs again with any of the subsequent three mounts, action shall be taken to make correction until three gun mounts in one lot have successfully completed tests in conformance to specified requirements Only after satisfactorily proving the requirements specified herein, will subsequent gun mount tests (production proof testing) be authorized.

4.8.1.1 *Gymnastication test.* To determine conformance to 3.5.2.1 and 3.5.2.1.1 the following shall be accomplished:

(a) The gun, gun mount recoil mechanism assembly shall be mounted on a gymnasticator and the travel limit switch contacts set to maximum recoil travel limits established for the particular mount being tested.

(b) Elevate the mechanism until the gun is as maximum elevation and gymnasticate (exercise) 20 times by moving the gun rearward the distance indicated in step "a" and then releasing it, permitting it to return to the "in battery" position under power of the recoil spring. After each gymnastication, the assembly shall be checked for time required to return to in battery position.

(c) Depress the assembly until gun is at zero degrees elevation and gymnasticate twenty times as above. After each gymnastication the assembly shall be checked for return cycle time.

(d) Depress the assembly to maximun degrees specified on applicable drawings and gymnasticate twenty times as above. After each gymnastication the assembly shall be checked for the return cycle time.

4.8.2 *Hydro-spring static pressure application test.* To determine conformance to 8.6.2.2, the recoil mechanism with the cannon in the assembly shall be statically pressure tested. The replenisher shall be disconnected from the recoil mechanism and the threaded port of the recoil mechanism connected to a pressure source containing specified hydraulic oil. The pressure source shall be capable of supplying a pressure for the specific assembly.

4.8.3 Charging hydro-pneumatic test. To determine conformance to 8.5.2.3, the nitrogen chamber shall be checked for trapped air.

4.8.4 *Provisions for proof-firing check.* To determine conformance to 3.5.3, the cannon and recoil mechanism shall be checked for damage to assembly after proof-firing.

4.8.5 *After proof-firing check.* To determine conformance to 3.5, 3.5.1, 3.6.3.1, 3.5.8.1.1 and 3.5.8.1.2, the first production gun mount and mechanism assemblies shall be checked as follows:

(a) The gun mounts and recoil mechanism assemblies shall be disassembled at the Government proving ground for visual and dimensional check

(b) Particular attention shall be exercised in checking the dimensions and conditions of the following critical parts:

1. Cylinders, cradles or housings.
2. Recoil dash pots (Orifice Dimensions) — cylinders, cradles or sleeves.
3. Recoil Pistons and rods.
4. Counter-recoil dash pots (Orifice Dimensions) — buffer pins, sleeve or buffor assemblies.
5. Retainers (Followers), bearings.
6. Replenisher assemblies, reservoirs — housings, functional or non functional parts.
7. Recoil spring(s) —load deflection characteristics.
8. Electrical firing circuits, linkage or contacts.
9. Breech operating mecha-

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nisms and related actuating devices.
10. Percussion firing mechanisms and related actuating linkage.
11. Recoil spring bearing seats.
12. All oil seals.
13. Relief valves, pressure pumps, equilibration check valves (recoil cylinders)

4.8.6 *Production gun mount assembly quality test.* To determine conformance to 3.5.4, each production gun mount assembly shall be gymnasticated as specified in 4.8.1.1 and proof-fired as specified in table II and checked for performance requirements of specific assembly being tested.

4.8.7 *Breech operation tests.*

4.8.7.1 *Sliding breech type.* To determine conformance to 3.5.4.1, the breech operating cam and extractors shall be observed for functional requirements, during the counter-recoil action.

4.8.7.2 *Automatic breech mechanism operation test.* To determine conformance to 3.5.4.2, during operation, the breech mechanism shall be checked for functional requirements.

4.8.8 *Adapter and collar check.* To determine conformance to 3.5.4.3 the adapters and collars shall be checked for proper torque and locking.

4.8.9 *Leakage test.* To determine conformance to 3.5.5, during and after the firing the gun shall be checked for leaks.

4.8.10 *Chamber pressure and muzzle velocity test.* To determine conformance to 3.5.6, the gun shall be loaded and fired in accordance with table I, chamber pressure and muzzle velocity measured and recorded.

4.8.11 *Recoil travel test.* To determine conformance to 3.5.7, the recoil travel shall be measured during proof-firing, and shall not exceed the specified length of travel.

4.8.12 *Recoil cycle time test.* To determine conformance to 3.5.7.1, the recoil cycle time shall be measured during proof-firing, and shall not exceed the time specified.

4.8.13 *Operation, complete recoil cycle check.* To determine conformance to 3.5.8, during operation of recoil mechanism the recoil cycle shall be checked for functional requirements.

4.8.14 *Safety devices check.* To determine conformance to 3.5.9, the safety devices shall be operated and checked for functional requirements.

4.8.15 *Electrical firing system check.* To determine conformance to 3.5.10 and 3.5.10.1, the electrical firing system and components shall be energized, and checcked for continuity of the electrical supply.

4.8.16 *Linkage assembly for percussion firing check.* To determine conformance to 3.5.11, the linkage assembly shall be operated and checked for functional requirements and linkage headspace.

4.9 **Failure.** If a gun mount fails to pass any test of first production or production tests, the Government inspector shall stop acceptance of subsequent fun mounts until objective evidence has been provided by the contractor that corrective action has been taken. any mount returned to contractor's plant after failure of a proving ground test shall be disassembled and reassembled only in the presence of the Government inspector. The Government inspector shall conduct tests of any one mount he selects to undergo testing in conformance with 4.8.1. Continuation of tests of any subsequent gun mounts will be discontinued until the provisions set forth in 4.8.1 and content herewith are acknowledged and endorsed by the Government inspector.

4.10 **Parts and components.** Parts, components and assemblies shall be inspected in accordance with supplemental quality assurance provisions listed on SQAP Index applicable to specified gun mount.

4.11 **Sampling for packaging and packing.**

4.11.1 **Lot formation.** A lot shall consist of all packs, packed level A (see 5.4.1), from 1 manufacturer, from an identifiable production period, submitted at one time for acceptance.

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4.11.2 *Sampling for accceptance examination.* Samples shall be selected at random from each lot in accordance with Standard MIL-STD-105.

4.12 **Examination inspection for packaging and packing.**

4.12.1 *Visual, dimensional, and primary functional examination.* Samples selected in accordance with 4.11.2, shall be examined for the following acceptance quality levels (AQL) on the basis of percent defective (see Standard MIL-STD-105):

Classification	*AQL*
Major	1.0
Minor	2.5

4.12.2 *Classification of defects.* For examination purposes, defects shall be classified as follows:

Major defect (1.0 AQL)	*Examination method*
Incorrect marking (see 5.5)	Visual
Illegible marking (see 5.5)	Visual

Major defect (2.5 AQL)	*Examination method*
Improper closure (see 5.4)	Visual

4.13 **Tests for packaging and packing.**

4.13.1 *Control tests.*

4.13.1.1 *Frequency.* Samples, prepared in accordance with level A, shall be selected at a rate of 2 per month or 4 in each 400 produced except that not more than 4 shall be selected in any give 30 day period. Samples shall be examined for the defects in 4.12.2 and subjected to the tests in 4.14.1 and 4.14.2.

4.13.1.2 *Test failure.* If a sample fails to pass a test specified herein, the Government inspector may stop acceptance of subsequent lots until objective evidence has proved that corrective action has been taken to eliminate the condition which caused rejection.

4.14 **Test procedure for packaging and packing.**

4.14.1 *Packaging test.* To determine conformnance to 5.3.1.3, the unit package shall be subjected to the tests specified for method III of Specification MIL-P-116.

4.14.2 *Packing test.* To determine conformance to 45.4.1, the pack shall be subjected to the interior packing test of Specification JAN-P-100.

5. PREPARATION FOR DELIVERY

5.1 **Disassembly.** Disassembly shall be accomplished only as necesary to safeguard parts from damage and to provide required reduction in cube. Wherever practicable, bolts, nuts, screws, pins, and washers shall be reinstalled on one of the mating parts and secured.

5.2 **Matchmarking.** When necessary to facilitate reassembly, removed parts shall be matchmarked. Tags used for matchmarking shall conform to type A of Specification UU-T-81. The marked tags shall be oversprayed with compound conforming to Specification MIL-V-13811 and secured to the parts.

5.3 **Preservation and packaging.**

5.3.1 *Level A.*

5.3.1.1 *Cleaning and drying.* Gun mounts shall be cleaned in accordance with process C-1 of Specification MIL-P-116, and dried by the most applicable procedure specified therein.

5.3.1.2 *Preservative application.* Moving parts, electrical contacts, and openings of electrical items shall be covered with tape conforming to Specification PPP-T-60. Electrical items shall then be sprayed with compound conforming to Specification MIL-V-13811. All exposed unpainted metal surfaces shall be coated with compound conforming to type P-4 of Specification MIL-P-116.

5.3.1.3 *Unit package.* All areas having a preservative application (see 5.3.1.2) and all small openings shall be covered with barrier material conforming to grade C of Specification MIL-B-121. Barrier material shall be secured with tape conforming to Specification PPP-T-60. Entire gun mount shall then be covered with wrappping paper conforming to Specification UU-P-271. Barrier material shall be secured with tape conforming to Specification PPP-T-60, but shall not be sealed. When tested as specified in 4.14.1, the unit package shall meet the requirement of method IIII of Specification MIl-P-116.

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5.3.14 *Disassembled items,* Disassem bled items (see 5.1) shall be preserved and unit packaged in accordance with the applicable requirements of Specification MIL-P-11443 and the packaged item shall be secured. with the shipping container (see 5.4.1).

5.3.2 *Level C.* Unless otherwise Specified (see 6.2), gun mounts shall be preserved and units packaged as specified in 5.8.1.1 and 6.3.1.9 except that all exposed, unpainted metal surfaces shall be coated with compound conforming to type P-6 of Specification MIL-P-116 and barrier material shall conform to grade A of Specification MIL-B-121. This level shall be used when shipping from contractor to test facility, and return.

5.4 **Packing.**

5.4.1 *Level A.* When net load will not exceed 3,000 pounds, gun mount shall be placed in a container conforming to Specification MIL-C-104. For net load exceeding 3,000 pounds, gun mount shall be placed in a container conforming to Specification MIL-C-104. When tested as specified in 4.14.2 the exterior pack shall meet the requirements of the interior packing test of Specification JAN-P-100.

5.4.1.1 *Blocking and bracing.* Gun mounts shall be blocked, braced, and anchored within the container in accordance with Specification JAN-P-100.

5.4.2 *Level C.* Gun mounts shall be packed in acordance with 5.4.1 through 5.4.1.1.

5.5 **Marking.** In addition to specified special marking (see 6.2) containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 **Intended use.** The gun mounts covered by this specification are intended for use in combat vehicles.

6.2 **Ordering data.** Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and class of gun mount required (see 1.2 and 6.4).
(c) Reproduction gun mount if required (see 3.1).
(d) Special aids if required (see 3.5).
(e) Any special marking required (see 5..5) .
(f) Type of packaging required (ace 5.3.2).
(g) Specify air cpmdotopmomg for assembly area (see 6.5).
(h) Approval of simulated vehicle application (see 6.6).

6.3 **Responsibility.** Responsibility and liability of the combination gun mount manufacturer or contractor shall continue until the gun mount has been declared acceptable by the proof-firing activity.

6.4 **Options.** Selection of the recoil system is governed by the characteristics of the weapon, such as size, purpose, rate of fire, and range of elevation. Option as to whether the mechanism shall be independent or dependent variable or constant recoil stroke, floating piston or direct contact, internal or external buffer, all are strongly influenced by basic factors such as recoil force and distance. space available, stability, and ground clearance.

6.5 **Assembly areas.** Assembly areas, special provisions, and equipment should be provided by the contractor. The assembly areas should be air conditioned using air conditioning unit with a 50 micron filter. The area temperature should be maintained between 68° to 72°F. The air flow rate per cubic inch, per minute should be such as to maintain area air pressure at between .76 and 1.5 inches of water. Relative humidity should not exceed 40-50 percent. Areas should be clean and free of dust, lint, and abrasive materials that may become entrapped in the assembled mechanisms. No grinding, filing, scraping, or any operation producing abrasive dust should be performed in the assembly area.

6.6 **Tank automotive items.** For Tank Automotive items, responsibility for approval of the simulated vehicle application rests with

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Commanding General, U.S. Army Ordnance Tank Automotive Command, 1501 Beard St., Detroit 9, Mich. Attn: ORDMC-REC.

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